Class 10: Structural Bioinformatics (Pt. 1)

Andrew Duong (PID: A18129684)

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THE PDB database

The Protein Data Base (PDB) is the main repository of biomolecular structure data. Let's see what is in it:

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

94% of structures are involved in both.

> Q2: What proportion of structures in the PDB are protein?

97% of structures are proteins

```
stats <- read.csv("pdb_stats.csv", row.name = 1)
stats
```

```
                        X.ray    EM   NMR Integrative Multiple.methods Neutron
Protein (only)         178795 21825 12773         343              226      84
Protein/Oligosaccharide  10363  3564    34           8               11       1
Protein/NA                9106  6335   287          24                7       0
Nucleic acid (only)       3132   221  1566           3               15       3
Other                      175    25    33           4                0       0
```

```
Oligosaccharide (only)      11     0     6           0                 1       0
                        Other  Total
Protein (only)             32 214078
Protein/Oligosaccharide     0  13981
Protein/NA                  0  15759
Nucleic acid (only)         1   4941
Other                       0    237
Oligosaccharide (only)      4     22
```

```r
stats/sum(stats$Total) * 100
```

```
                              X.ray         EM         NMR Integrative
Protein (only)          71.800030520 8.76442667 5.129348079 0.137741047
Protein/Oligosaccharide  4.161546555 1.43122184 0.013653631 0.003212619
Protein/NA               3.656763768 2.54399280 0.115252713 0.009637858
Nucleic acid (only)      1.257740404 0.08874860 0.628870202 0.001204732
Other                    0.070276044 0.01003943 0.013252054 0.001606310
Oligosaccharide (only)   0.004417351 0.00000000 0.002409464 0.000000000
                        Multiple.methods      Neutron        Other        Total
Protein (only)              0.0907564915 0.0337325013 0.0128504767 85.968885783
Protein/Oligosaccharide     0.0044173514 0.0004015774 0.0000000000  5.614453574
Protein/NA                  0.0028110418 0.0000000000 0.0000000000  6.328458184
Nucleic acid (only)         0.0060236609 0.0012047322 0.0004015774  1.984193914
Other                       0.0000000000 0.0000000000 0.0000000000  0.095173843
Oligosaccharide (only)      0.0004015774 0.0000000000 0.0016063096  0.008834703
```

```r
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits =2)
```

```
           X.ray               EM              NMR      Integrative
            0.81             0.13             0.06             0.00
Multiple.methods          Neutron            Other            Total
            0.00             0.00             0.00             1.00
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

4940 structures

What is the total number of entries in the PDB?

```
n.sums["Total"]
```

```
 Total
249018
```

Using Molstar

We can use the main Molstar viewer online:

> Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing the inhibitor (ligand) in spacefill.

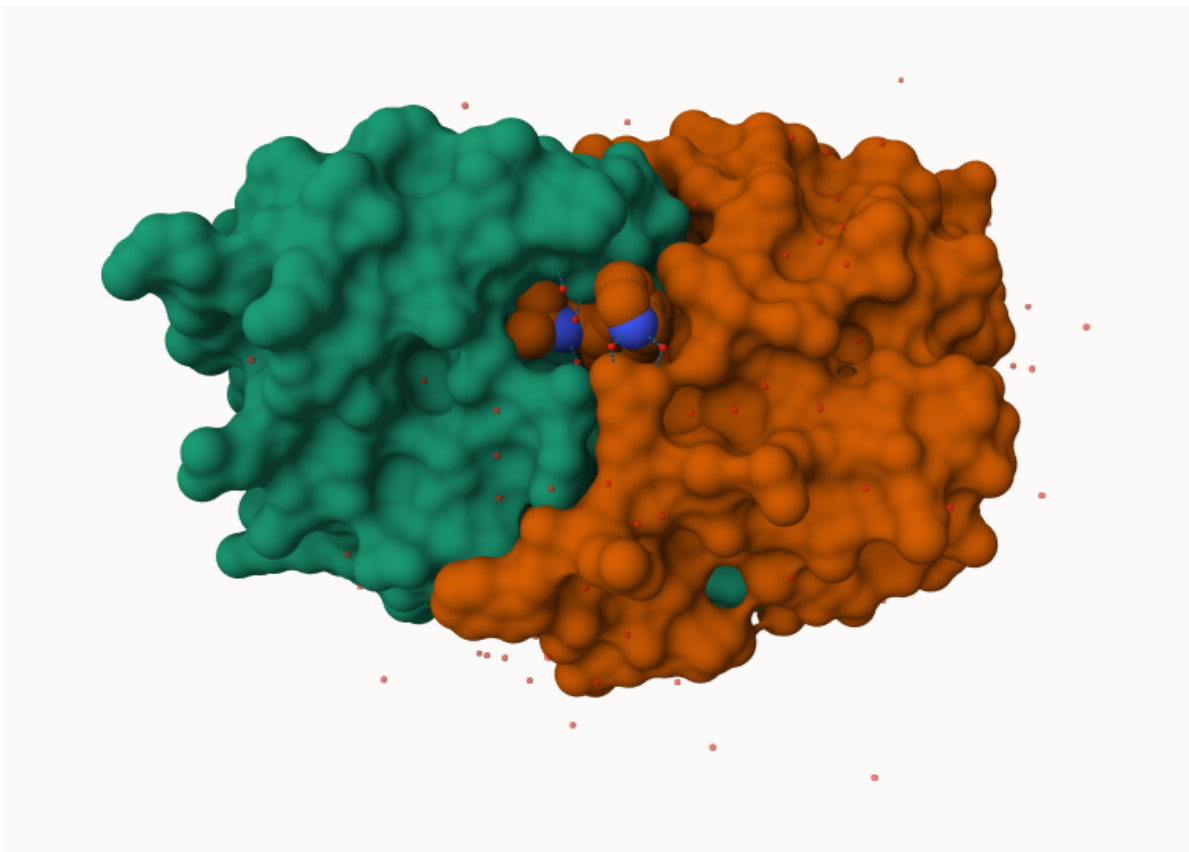
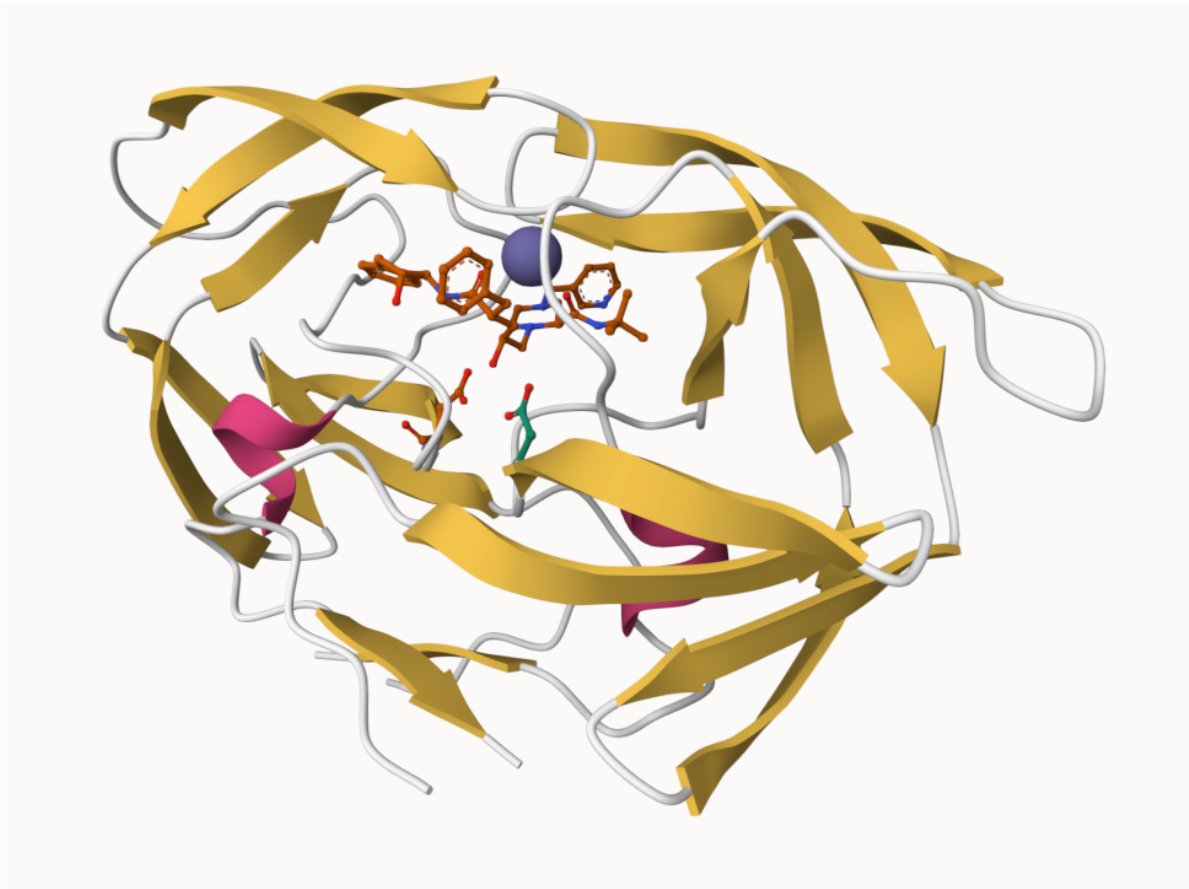


Figure 1: First view of HIV-Pr dimer with bound inhibitor

> Q. One final image showing catalytic APS 25 and the all important active site water molecule as spacefill.



The Bio3D package for structural bioinformatics

```
library(bio3d)

hiv <- read.pdb("1HSG")
```

```
  Note: Accessing on-line PDB file
```

```
hiv
```

```
 Call:  read.pdb(file = "1HSG")

   Total Models#: 1
```

```
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(hiv$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
pdbseq(hiv)
```

```
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K"
 21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G"
 41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
```

```
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D"
 61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
 81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99   1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
  2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
 22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40  41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
 42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
 62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
 82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"
```

Let's try out the new **bio3dview** package that is not yet on CRAN.We can use the **remotes** package to install any R package from GitHub.

Quick Viewing of PDBs

```r
library(bio3dview)

sele <- atom.select(hiv, resno = 25)
# view.pdb(hiv, backgroundColor = "teal",
#         highlight = sele,
#         highlight.style = "spacefill")
```

Prediction of Protein flexibility

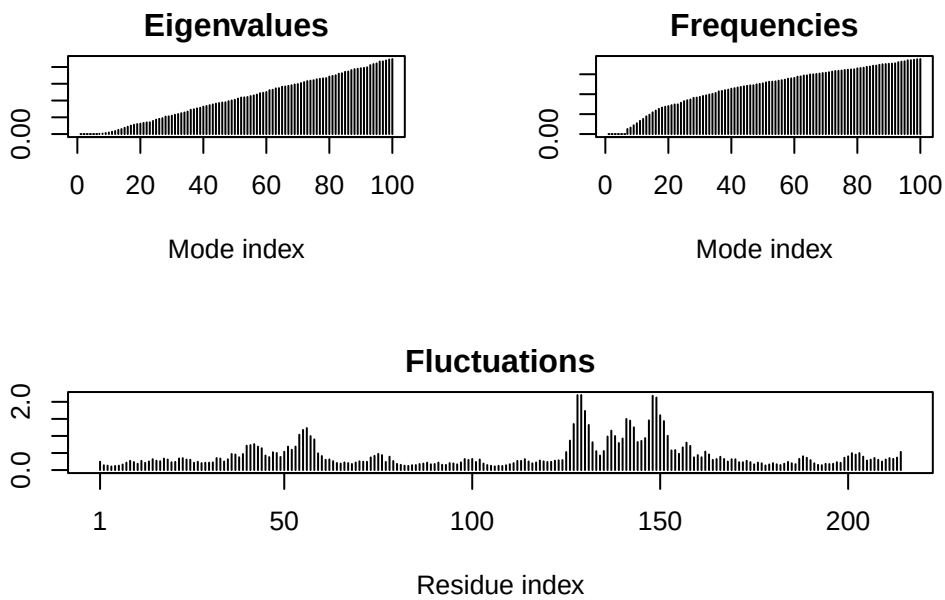
```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

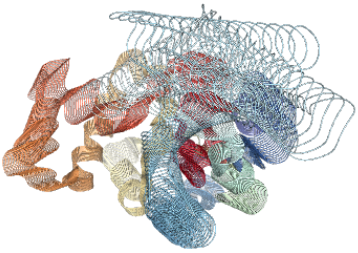
```r
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.01 seconds.
 Diagonalizing Hessian...   Done in 0.17 seconds.
```

```r
plot(m)
```



Write out our results as a wee trajectory movie:

```r
mktrj(m, file = "results.pdb")
```

```r
view.nma(m)
```